

⊕ SUBNET MAGAZINE

SUBNET PROFILE · Issue 011, Storage on chain

Alphanomics: A new incentive model by SN75 Hippius

A close read of Hippius, the storage subnet quietly pricing itself 2,386× cheaper than AWS, Google Cloud, Azure, and Filecoin.



⊕ 2025-09-27 · filed by Subnet Magazine

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Model by SN75 Hippius

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In this article, we dive into Subnet 75, @hippius_subnet, a globally distributed, transparent, decentralized cloud storage and virtual machine network. Hippius is currently the only successful subnet in the PBittensor ecosystem focused on scaling cloud storage, positioning itself to compete against centralized and decentralized heavyweights like PAWScloud, PGoogleCloud, PAzure, and PFilecoin in a highly competitive, continuously growing \$900 billion industry. Priced against its centralized competitors, Hippius is 2,386 times cheaper—I'd say they're off to a good start.

The approach Hippius takes to scale seems to be out of the ordinary compared to most subnets. From creating its own blockchain bridge to smart contract-pegged Alpha, Hippius is striving to set a new standard within the Bittensor ecosystem.

According to one of the founders, @mogmachine, Hippius was created to address the issue of storage on Bittensor, noting that most subnets offering compute services focus on power and GPU distribution, but there is no successful memory bank in the ecosystem. There have been subnets in the past that attempted decentralized storage on Bittensor but failed. There has always been, and will continue to be, a need for decentralized storage. As the amount of data grows exponentially from the training and development of AI models, it becomes increasingly important to maintain control over your data rather than leaving it in the

hands of centralized providers that monetize it and exert excessive influence over personal information.

How Does Hippus Secure and Distribute Data?

One of the first questions many may ask about Hippus is how it secures data. One of the first questions many may ask about Hippus is how it secures data safely and efficiently. The backbone of Hippus's data

securitization is the InterPlanetary File System (IPFS), a digital infrastructure technology developed by PJuanBenet of PFilecoin and PProtocolLabs. In simple terms, IPFS is a method for storing and sharing data without relying on a central server. Instead of using mutable web addresses like URLs, IPFS identifies files based on a unique fingerprint derived from their content—a cryptographic hash that remains constant. Data on IPFS can be encrypted or public, depending on the preferences of the key owner or developer using the technology.

Hippus integrates IPFS directly into its subnet network, leveraging it as the primary protocol for decentralized file pinning and distribution.

When data is uploaded to Hippus, it is first encrypted end-to-end on the user's device using a robust cryptographic standard such as AES-256, ensuring that only the user holds the keys for decryption. The data is then broken down into smaller chunks and distributed across its global peer-to-peer network of volunteer nodes for redundancy and resilience. A notable feature of Hippus is that users can locate the nodes holding their data through the chain's block explorer on Hippus's new Substrate blockchain.

Hippus Blockchain and hAlpha

Hippus has introduced an innovative concept by deploying a Substrate blockchain directly within its subnet. A key objective behind this is to explore methods for driving value back to its Alpha token. Hippus has been working on unlocking value by deploying its WASM smart contracts natively on the Bittensor main chain. If this proposal is approved by POpentensor, it would enable Hippus to establish a fully native bridge from its chain to Bittensor, avoiding the use of EVM and enhancing robustness. This bridge would facilitate the swapping of Alpha to hAlpha, which settles on the Hippus chain. hAlpha is paired one-to-one with its native Alpha on Bittensor, meaning the value on the Hippus chain is directly linked to the Bittensor-native Alpha token. Essentially, hAlpha serves as a 1:1 mirror of the Bittensor-native Hippus Alpha token on the Hippus blockchain, enabling seamless operations without typical cross-chain complexities. Users can bridge their Alpha via this two-way Hippus bridge to obtain hAlpha. This architecture not only streamlines interactions but also represents a significant advancement in retaining and circulating value within the ecosystem, particularly as Hippus scales to meet real-world storage demands.

hAlpha: Hippus Staking and Miner Lockups

To explore the token mechanics further, consider staking and miner lockups, as these elements drive key incentives. Holders of Hippus Alpha can bridge their tokens to hAlpha and stake them on the Hippus chain in the same manner as on Bittensor, but with additional benefits including distributed dividends, governance participation, and revenue shares that enhance the appeal beyond the native chain.

For miners, the requirements are more stringent. To participate, miners must bridge and lock up hAlpha tokens as collateral for the storage they provide on the network. As Hippius expands, driven by the rapid growth in AI-related data, more hAlpha becomes locked, which influences the supply and demand dynamics of the original Alpha token while strengthening the value proposition for hAlpha holders. This creates a reinforcing cycle: increased locked tokens enhance secure storage capacity, attracting more users and leading to further lockups. The storage supply versus Alpha value chart illustrates this correlation, showing consistent growth in the collateral system as the network scales. Miners who lock up Alpha for hAlpha forfeit the dividends they would otherwise earn on the Bittensor side for those tokens. However, this forfeited yield is not lost; it is pooled and automatically distributed to all other hAlpha stakers on the Hippius network. As a result, staking hAlpha offers potentially higher returns than native staking on Bittensor. This miner lockup mechanism transforms individual forfeitures into collective benefits, fostering sustained participation across the subnet.

Distributed Revenue: How Usage Generates Value

Revenue generation is central to sustaining a decentralized system like Hippius. Product revenue, whether in fiat or cryptocurrency, flows through the bridge, is converted to hAlpha, and remains locked until a customer utilizes the storage or compute services. During this lockup period, the funds do not earn dividends themselves but contribute to enlarging the overall dividend pool for hAlpha stakers, thereby increasing payouts network-wide and creating a flywheel effect where greater usage elevates staker rewards.

Upon usage, such as data retrieval or virtual machine execution, the hAlpha credits are allocated to affiliates, referrals, and other predefined channels

on-chain, with governance determining the distribution to ensure fairness. This includes shares for referrals that onboard new users, affiliates for partnerships, and a treasury for ecosystem grants or community-approved initiatives. All allocations are transparent and verifiable.

To complete the cycle, dividends from miner-locked hAlpha and revenue-locked funds generate additional dividends, which are then distributed to every hAlpha staker on the Hippius blockchain. This structure makes bridging and staking hAlpha an compelling option, allowing holders to benefit from network growth and actual usage. The architecture overview depicts this flow: Bittensor feeds into Hippius miners, channels through the Hippius chain to a central dividend pool, and delivers enhanced yields to stakers (combining native Bittensor dividends with Hippius additions). Revenue is governed into categories like referrals, affiliates, and treasury, completing the loop. As of late September 2025, with Hippius securing over-the-counter investments such as the \$100K from PDSVFund and providing regular updates on X, this revenue model demonstrates long-term viability.

Verified On-Chain

A distinguishing feature of Hippius in the decentralized storage landscape is its commitment to on-chain verification, eliminating the need for unverified trust. The Substrate blockchain logs every upload, chunk distribution, node pinning, and revenue flow in immutable records. Users can audit their data's path via the block explorer, tracing IPFS hashes to specific volunteer nodes holding encrypted chunks. Bridge

transactions from Alpha to hAlpha are supported by on-chain proofs that maintain the 1:1 peg, with WASM contracts (pending POpentensor approval) ensuring native integration and resilience over EVM-based alternatives. Dividend distributions, including miner forfeitures, revenue locking and unlocking, and governance splits, are all timestamped and auditable. In contrast to centralized clouds where data management can lack transparency, this on-chain layer provides a reliable foundation for user confidence.

Upcoming Developments

Hippius continues to advance its capabilities, with 2025 marking significant progress. Following in-depth launch sessions in June and positive coverage from influencers in August and September, highlighting Subnet 75 as a foundational element of Bittensor, the team is focusing on expanding virtual machine functionality to support intensive AI workloads. Enhancements to node incentives aim to attract more global participants, reducing latency for real-time applications. Governance tools are also being refined, enabling stakers to influence revenue distributions or bridge features. With the Bittensor mainnet upgrades that began on September 26, 2025, which introduced improved incentive mechanisms for subnet performance, Hippius is well-positioned to accelerate its growth. A recent Revenue Search episode

(hosted by @siamkidd and PMarkCreaser) featuring @mogmachine and @mast3rdubs discussed miner optimizations, including potential automated lockup adjustments based on demand. Should the WASM deployment receive approval, the native bridge could launch sometime in Q4, facilitating greater interoperability across Bittensor subnets.

Roadmap

Looking forward, Hippius's roadmap outlines a path toward leadership in decentralized cloud services. In the short term (Q4 2025), priorities include securing POpentensor approval for the WASM bridge, deploying beta governance dashboards for hAlpha holders, and achieving 1PB of verified storage capacity through miner incentives. In the mid-term (Q1-Q2 2026), the focus shifts to integrations with other Bittensor subnets for combined compute-storage solutions, introducing fiat on-ramps for revenue, and conducting third-party audits of the on-chain verification system. The long-term vision is to serve as the primary backend for AI data pipelines, collaborating with key players in the ecosystem while maintaining costs 2,386 times below those of major clouds. With subnet growth increasing 50% in Q2 2025 and new investments from funds like PStillcoreCap (with @markjeffrey and @jason at the helm) targeting Alpha tokens, Hippius is poised to capitalize on Bittensor's expansion. Ultimately, it aims to establish the data infrastructure for future AI applications through systematic, decentralized pinning.

Slides provided by: @mogmachine

Article submitted by: PRondo_ina_Condo Laron Campbell and @shifa_yyz Shifa Abbas

Simplifying Bittensor subnet ecosystem through accessible, insightful articles. #TAO